



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Using Blockchain for Traceability, Fraud Prevention, and Smart Contracts.

ASSIGNMENT 1 - TECHNICAL REPORT

Course : Enterprise Systems Design & Modelling (SECP3744)

Semester/Session : 1-2025/2026

Section : 02

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1.0 Introduction

The blockchain system marks the inception of a revolutionary shift within the realm of business operations. Although it initially attracted popularity as an aspect of digital currencies such as Bitcoin and Ethereum, the principles of blockchain can be applied much more effectively than digital currencies alone. For example, blockchain functions in contrast to the central database system as it involves a collective management approach by various users such that once information has been placed on the database, it cannot be manipulated. This distinct trait of blockchain makes it extremely useful in business settings where trust, integrity, and accountability play important roles

In addition, the integration of blockchain with smart contracts, which are automatically executed contracts programmed on the blockchain itself, has brought in a new dimension of automation in business operations. The smart contracts automatically enforce rules and execute business operations as and when certain conditions arrive. The purpose of writing this paper is to explore the mechanisms that work to improve the traceability, prevent fraud, and automate business processes through blockchain. The author of the paper believes that by understanding such mechanisms, organizations will be in a better position to comprehend blockchain solutions.

2.0 Literature Review

The blockchain based on DLT records transactions on various nodes. This provides data integrity without having a central controlling entity. The structure of blockchain forms an immutable records database since it connects a new block to the previous ones using cryptography. The blockchain concept was proposed by Nakamoto in 2008.

Current research emphasizes the increasing use of blockchain applications beyond cryptocurrency. The role of blockchain in supply chain management as discussed by Kshetri (2018) and Saberi et al. (2019) in preventing counterfeiting of goods such as food and medications cannot be overstated. The important aspect of blockchain technology that stands to transform the world of business as we know it is smart contracts proposed by Szabo (1996).

Enterprise adoption has accelerated with platforms like Ethereum and Hyperledger Fabric, which enable building permissioned networks (Deloitte, 2023; IBM, 2024). However, challenges persist, including scalability, interoperability, and regulatory uncertainty (Crosby et al., 2016; Yli-Huumo et al., 2018). Despite these barriers, blockchain's potential for ensuring data trustworthiness and automating multi-party transactions continues to drive enterprise innovation.

3.0 Analysis and Discussion

3.1 How Blockchain for traceability works

Blockchain for traceability works by recording every transaction or movement of a product as a digital “block” in a shared ledger that is distributed across many computers. Each block is time-stamped and linked to the previous one, creating an unchangeable chain of records. This means that all participants, such as manufacturers, suppliers, and consumers, can access the same verified information without relying on a single central authority. As a result, blockchain ensures data transparency, improves trust among stakeholders, and reduces the risk of fraud or counterfeit products.

The **benefits** of blockchain in traceability are significantly used by major companies such as (Deloitte US, 2025). It enhances transparency by allowing all parties to view verified product data, strengthens security through cryptographic protection, and increases efficiency by automating updates with smart contracts. Industries such as food, pharmaceuticals, and luxury goods are already using blockchain to track products from origin to consumer. For instance, Walmart uses IBM’s Food Trust blockchain to trace the source of food items, allowing contaminated products to be identified within seconds. Similarly, companies like LVMH use blockchain to verify the authenticity of luxury products (Tian, 2021).

However, blockchain traceability faces several **limitations**. The accuracy of information depends on correct data entry. If false data is entered, it remains permanently stored. Implementation can also be costly and technically complex, especially for smaller businesses (Ioscm, 2022). Despite these issues, blockchain significantly improves transparency, accountability, and trust within supply chains by ensuring every transaction is securely recorded and verifiable.

3.2 How Blockchain for fraud prevention works

The main use of the blockchain technology in fraud protection appears in its properties of decentralization, unalterability, and openness (NIST, 2018). Contrary to traditional storage systems, for example, a central authority manages the system, creating a single “point of failure” (Rapid Innovation, 2024). Thus, instead, the “blockchain is a decentralized ledger, whereby a copy of the same information is shared by several participants in a given network” (IBM, 2023). This feature, in turn, renders it impossible to hack, since any form or type of information would need the majority’s consent (Yaga et al., 2018; Rapid Innovation, 2024). Moreover, the information for each transaction is stored in “blocks,” linked through “cryptographic rules in a ‘chain,’” making it unalterable (SoluLab, 2024).

It, therefore, appears that the main advantage or use for fraud detection using the blockchain technology is its attributes of “decentralization, unalterability, and openness” (NIST, 2018).

The first **benefit** that can be derived from that technology is that it represents a paradigm shift from reactive fraud analysis to preventive fraud analysis. This tamper-proof feature guarantees the accuracy of the information (ResearchGate, 2025) and also helps in making a single, undeniably true source (IBM, 2023). This helps each and every authorized party access a clearly defined, time-stamped audit path that helps identify assets and financial transactions (World Bank Blogs, 2025; Masverse, 2024) so that any attempt for fraud, manipulation, or counterfeiting is made highly expensive by the use of blockchain technology, ultimately leading to increasing confidence in stakeholders for valuable financial transactions.

Although it provides strong security, the application of the concept of blockchain technology for fraud protection has **limitations**, mainly associated with governance and adoption. The process requires a consensus from the majority of a decentralized system for any modifications or validation in terms of information, thereby making operations in such a system sometimes less efficient than in a centralized system. The integration process for a new system that uses blockchain technology can also be expensive, requiring experts in blockchain technology and designing new technology infrastructures that can support it.

3.3 How Blockchain automation through smart contract works

Smart contracts can be defined as a self-executing code that runs on a Blockchain. Smart contracts can also be identified as code containers that execute the terms of a real-world Contract in the digital world (Investopedia, 2025). Smart contracts can also act as third-party replacement for the parties involved in a contract since it executes its code, which is automatically broadcasted and validated by the nodes present in the Blockchain (Taherdoost, 2023). Smart contracts enable untrusted parties to conduct transactions without requiring interaction from the concerned parties.

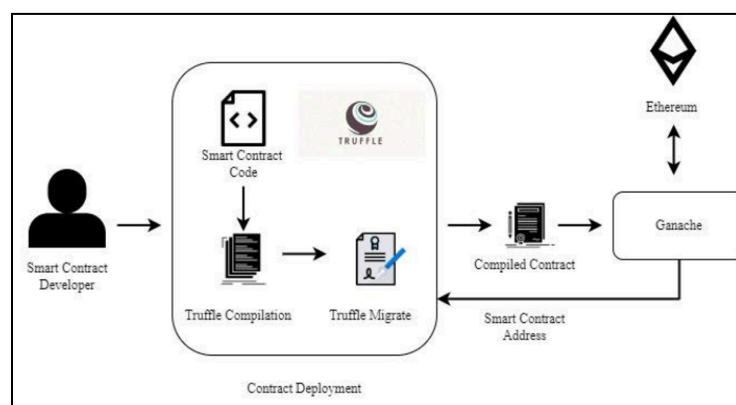


Figure 3.3: System Architecture Diagram illustrating the deployment flow of a Smart Contract onto the Ethereum blockchain using the Truffle framework.

As illustrated in the example above, the architecture that supports the automation of smart contracts requires a multi-step deployment process, whereby the first step is carried out by the Smart Contract Developer, creating the code (Ahamed, N Nasurudeen. 2023). The code is then compiled through Truffle Compilation, when it is interpreted from human-readable code to the machine-readable form of the Compiled Contract. The Truffle Migrate process then follows, facilitating the deployment, utilizing a local testing environment such as Ganache to mimic the production Ethereum network. The end process then gives a Smart Contract Address, acting as the interface through which third-party applications can use the automated logic defined by the Smart Contract (Ahamed, N Nasurudeen. 2023).

The use of smart contracts has **benefits** that contribute to the emergence of a trustless, automated business logic, leading to improved efficiencies and savings for operations. Smart contracts remove human interaction, facilitating faster transaction action (such as payments, transfers) triggered by checked conditions, offering near real-time settlement and governance through automation and the absence of middlemen, thereby reducing the risk level, achieving lower services, and improved efficiencies in contrast to traditional contracts (Taherdoost, 2023).

However, there can also arise a number of **negative impacts** from the concept of smart contracts. The codes for executing smart contracts are immutable once they are placed upon the blockchain, and any flaws present in the code will also remain so. This can lead to unforeseen problems that can arise from such immutability. Moreover, although it does away with the middlemen, the deterministic code can give rise to problems for settling disputes that need a certain level of interpretation beyond the initial scope discussed in the code (The Cambridge Handbook of Smart Contracts, 2023).

3.3 Challenges and future predictions for blockchain implementation

Blockchain technology offers transparency, security, and decentralization, but its implementation still faces several significant challenges. One of the main issues is data accuracy. Although blockchain ensures that stored data cannot be changed, it cannot guarantee that the information entered is correct. This “garbage in, garbage out” problem means inaccurate input will remain permanently on the ledger. Another major challenge is high implementation cost, as developing and maintaining blockchain infrastructure requires advanced hardware, software, and skilled personnel. Small and medium-sized enterprises often struggle to afford these expenses (Ioscm, 2022).

Scalability also remains a concern since most blockchain systems can only process a limited number of transactions per second, leading to slower performance compared to traditional centralized databases. Additionally, energy consumption is an issue, especially for blockchains using the Proof-of-Work mechanism, which requires large computational power and electricity. Integration with existing systems is another difficulty, as organizations must adapt their workflows to fit blockchain frameworks. Finally, legal and regulatory uncertainty across different countries creates complications in cross-border blockchain applications, and privacy concerns arise due to the high transparency of blockchain records.

Looking to the future, blockchain adoption is expected to expand across industries such as finance, logistics, and healthcare as technology becomes more efficient and affordable. Governments are also likely to introduce **regulatory frameworks** to promote safe and standardized blockchain use. The integration of blockchain with **Artificial Intelligence (AI)** and the **Internet of Things (IoT)** will further enhance automation, data accuracy, and decision-making. Moreover, the shift toward **energy-efficient consensus models** like Proof-of-Stake will make blockchain more sustainable. Central Bank Digital Currencies (CBDCs) and cross-chain interoperability are also expected to drive global innovation.

4.0 Recommendation

Although blockchain offers transparency, decentralization, and security, its successful implementation requires addressing several major challenges. One of the key issues is ensuring data accuracy, as inaccurate information recorded on the blockchain cannot be altered. To overcome this, industries can adopt Internet of Things (IoT) sensors, QR codes, and automated data collection tools that directly capture verified information from the source, minimizing human error. Another challenge is the high cost of implementation, especially for small and medium-sized enterprises. This can be reduced by adopting Blockchain-as-a-Service (BaaS) platforms such as IBM or Oracle, which provide ready-made blockchain frameworks with lower setup and maintenance costs.

Scalability and energy consumption are also pressing concerns. Using layer-2 solutions or hybrid blockchain systems can improve transaction speed and network capacity. Meanwhile, transitioning from Proof-of-Work (PoW) to Proof-of-Stake (PoS) or Proof-of-Authority (PoA) mechanisms can significantly reduce energy usage while maintaining network security. Additionally, industries should focus on system integration by developing standardized APIs and interoperability protocols that allow blockchain to work seamlessly with existing enterprise systems such as ERP or CRM software.

To align blockchain technology with industry needs, organizations must emphasize collaboration, education, and standardization. Industry consortia, such as IBM Food Trust and MediLedger, show how shared blockchain networks can promote trust and consistency. Continuous training and research partnerships between companies, universities, and technology providers will also drive innovation and address emerging issues such as interoperability and sustainability.

5.0 Conclusion

Blockchain technology has evolved from its origins in cryptocurrency to become a transformative force in enterprise systems. Its decentralized, immutable, and transparent nature enables organizations to enhance traceability, prevent fraud, and automate operations through smart contracts. Across industries such as supply chain, finance, and healthcare, blockchain strengthens trust and accountability by providing a single source of verified truth among stakeholders. However, despite its benefits, adopting blockchain comes with challenges. High implementation costs, accuracy of data input, scalability issues, and unclear regulations still limit its widespread use. These obstacles point to the need for technological innovation and shared standards to ensure blockchain's long-term functionality and compatibility with existing systems.

Moving forward, the successful integration of blockchain will depend on how effectively organizations can align it with emerging technologies like IoT and AI, and how governments establish clear regulatory frameworks to guide its ethical and secure use. With continuous improvement, research, and cross-industry collaboration, blockchain has the potential to become a foundational technology for the next generation of transparent, efficient, and trusted digital ecosystems.

6.0 Appendices

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7.0 Executive Summary

This technical report analyzes the critical role of **blockchain technology** in modern enterprise systems. The focus is on how blockchain's core properties like **decentralization, immutability, and transparency**—directly address key challenges.

The report specifically investigates three key applications:

1. **Enhanced Traceability:** Providing a secure, end-to-end, and verifiable history of assets in the supply chain.
2. **Fraud Prevention:** Mitigating counterfeiting and falsification risks through unchangeable records.
3. **Operational Automation:** Streamlining business processes and contractual agreements using **smart contracts**.

The conclusion is that blockchain offers a robust solution for increasing enterprise security, trust, and efficiency, providing a foundation for highly transparent and automated business environments.

8.0 Logbook

Prompt	Date	Time	Verification Steps
How blockchain works in automation of smart contract	20/10/2025	9.43 PM	Finding the cited resources through google and aligning the content
Current development blockchain in industry and future expectation	22/10/2025	9.41 PM	Align it with the related research paper

